Embodied Stylistic Planning via Structured Reasoning and Execution Feedback

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***Abstract*—Enabling robots to translate high-level human intent into physically grounded action remains a central challenge in human–robot interaction. Although recent vision–language and large language models can generate structured plans from natural language, they lack embodiment awareness and often produce physically infeasible or perceptually misaligned behavior. This gap is particularly pronounced in expressive tasks, where users specify stylistic qualities (e.g., *"play this in jazz style"*) that must be realized through temporally precise, constraint-bound motor control. Robotic piano performance exemplifies this challenge, requiring high-frequency dexterous execution under strict hardware limitations. We present ELISE, a structured multi-stage VLM framework for style-conditioned dexterous skill synthesis under embodiment constraints. Rather than collapsing intent interpretation, hardware adaptation, and control generation into a single prompt, ELISE decomposes reasoning into sequential stages spanning musical intent analysis, embodiment-constrained planning, robot-API-level code generation, and iterative refinement using execution-derived feedback. This staged design enables explicit alignment between abstract stylistic intent and physically feasible execution while maintaining transparency across reasoning layers. We evaluate ELISE on a humanoid robot equipped with constrained dexterous hands, performing three songs under original, jazz, and blues styles. In a blind ranking study across 24 real-robot performances, the proposed framework significantly improves both melodic faithfulness and perceptual style alignment compared to direct single-prompt code generation.**

I. Introduction

As robots increasingly operate in human-centered environments, they are expected not only to execute tasks accurately but also to interpret and express high-level human intent. In everyday interaction, users rarely specify behavior in terms of joint trajectories or control parameters. Instead, they communicate goals in abstract, semantic, and stylistic terms—describing not only *what* should be done, but *how* it should be performed. Instructions such as *"make it more fluid," "play more energetically,"* or *"perform this in a jazz style"* encode qualitative expectations about timing, dynamics, and expressive character. Translating such underspecified and subjective descriptions into physically grounded, temporally precise robot actions requires reconciling human-level semantic intent with embodiment constraints, including kinematic reach, actuation bandwidth, and control latency. Achieving this alignment between abstract instruction and constraint-bound motor execution remains a central challenge in human–robot interaction (HRI).

Recent advances in vision–language and large language models (VLMs/LLMs) enable robots to generate structured plans and executable code from natural language instructions [1–4]. However, these models reason primarily in a symbolic space and lack intrinsic awareness of embodiment constraints such as kinematic reach, transition latency, and actuation limits [5–7]. Consequently, directly generated plans often violate hardware limits or produce behavior that is perceptually misaligned with the requested style. This gap between *expressive intent* and *physical execution* is especially pronounced in expressive, high-frequency dexterous tasks such as piano performance, where stylistic variation must be realized within strict hardware constraints.

We argue that bridging this gap requires structured reasoning that separates intent interpretation from embodiment grounding and control realization. Our key insight is that the mismatch between human-level style descriptions and robot-level motor execution can be mitigated by **decomposing reasoning into sequential stages** rather than collapsing the entire process into a single prompt. Each stage operates on a well-defined intermediate representation, preventing the language model from simultaneously reasoning about musical semantics, hardware limitations, and low-level API syntax.

We instantiate this idea in ELISE (Embodied Language for Intent-to-Skill Execution), a multi-stage LLM framework that maps a song title and desired playing style to executable robot control code. The pipeline consists of musical intent analysis, embodiment-constrained planning, robot-API-level code generation, and iterative refinement using execution-derived feedback. By explicitly modeling hardware constraints and grounding corrections in real execution outcomes, ELISE aligns high-level stylistic intent with physically feasible dexterous motion.

We evaluate ELISE on expressive robotic piano performance using a humanoid platform equipped with a dexterous hand operating under both hard constraints (thumb excluded, white keys only) and soft constraints (e.g., repositioning latency and actuation bandwidth). Piano playing serves as a rigorous HRI testbed: it demands high temporal precision, fine-grained velocity modulation, and stylistic variation that is directly perceptible to human observers. In a blind ranking study covering 24 real-robot performances across three songs and three stylistic conditions (original, jazz, and blues), participants consistently preferred ELISE-generated performances over direct single-prompt generation in terms of style alignment and coherence.

This work makes three primary contributions:

- We introduce ELISE, a structured multi-stage framework for language-conditioned stylistic planning under embodiment constraints, enabling expressive dexterous control without style-specific training data.
- We propose an execution-grounded refinement mechanism that leverages structured outcome feedback rather than corrective language or manually specified reward signals to iteratively improve alignment between plan and hardware execution.

- We provide empirical evidence through real-robot experiments and a blind user study demonstrating that staged reasoning significantly improves perceptual style alignment compared to direct LLM-based code generation in high-frequency dexterous tasks.

More broadly, our results suggest that structured language-to-action pipelines offer a principled path toward improving reliability, interpretability, and perceptual alignment in expressive HRI.

II. Related Work

Language as an Interface for Expressive Robot Control
In human–robot interaction (HRI), robot motion serves both functional and communicative roles. Variations in timing, velocity, and trajectory shape convey affect and intent, influencing human perceptions of competence and trust [8–10]. Prior work has demonstrated expressive modulation in gesture synthesis, locomotion, and socially aware manipulation [11–13], but stylistic behavior is often encoded through predefined parameters or learned embeddings rather than specified directly through natural language. Recent large language and vision–language models enable robots to translate language instructions into structured plans and executable control programs [1–3, 14, 15], often augmented with feedback for improved robustness [5, 6, 16]. Beyond discrete task execution, emerging work leverages language to generate reward functions for downstream optimization [17, 18], reason about socially appropriate behaviors given robot capabilities [19, 20], or combine language-generated rewards with preference learning to efficiently produce expressive locomotion [21]. These approaches demonstrate that language can guide perceptually meaningful and stylistically rich robot behaviors.

However, existing methods typically rely on learned policies, explicit reward shaping, or direct human preference signals to refine behavior. When feedback is incorporated, it often takes the form of additional language instructions, manually specified corrections, or scalar reward updates. In contrast, our setting considers high-frequency dexterous manipulation in which stylistic intent must be translated into explicit, embodiment-constrained action sequences and refined using only the outcome of execution. Rather than receiving corrective language or annotated error signals, the system observes structured execution results (e.g., MIDI transcriptions) and uses them to iteratively update its plan. This formulation emphasizes alignment through outcome-based grounding under strict physical constraints, treating language as an expressive control interface whose outputs must satisfy both perceptual intent and embodiment feasibility on real hardware.

Embodiment-Aware Planning and Constraint Grounding
Ensuring physical feasibility is a longstanding concern in robotics [22, 23]. In LLM-based robotic systems, hallucinated or underspecified outputs can translate into physically infeasible or unsafe actions [24, 25], motivating research on improving reliability without extensive task-specific fine-tuning. Existing approaches can be grouped into three broad categories. First, *prompt engineering* techniques—such as CoT, ReAct, and RAG—encourage more structured reasoning, sometimes augmented with simulator or human feedback [6, 26, 27]. Second, *run-time verification* and shielding frameworks translate safety constraints into formal specifications and filter potentially unsafe actions during execution [28, 29]. Third, *symbolic integrations* structure LLM outputs using representations such as PDDL or domain-specific grammars to facilitate constraint reasoning [1, 15, 30, 31]. While these methods improve reliability and structure, they are largely designed for well-defined task objectives. Classical constraint-aware planners optimize under explicit kinematic and dynamic limits, and stylistic variation—when considered—is typically encoded as a cost term or learned from demonstrations [9, 21, 32]. Such formulations assume a fixed objective structure and rarely address open-ended, language-specified style interacting with strict embodiment limits in high-frequency dexterous tasks.

We argue that constraint interpretation is task-dependent and often requires grounding in execution outcomes. ELISE embeds explicit embodiment priors within a staged language-based reasoning pipeline and refines plans using execution feedback, enabling expressive intent to be adapted to constrained hardware without manually specified safety rules, reward functions, or task-specific training.

III. Method

We present ELISE, a structured multi-stage reasoning framework for aligning high-level stylistic intent with physically feasible dexterous robot execution. Rather than using a single prompt to jointly interpret intent, enforce hardware constraints, and generate control code, ELISE decomposes reasoning into sequential stages operating at distinct levels of abstraction. Each stage produces a structured intermediate representation that is explicitly passed to the next, enabling clear separation between perceptual reasoning, embodiment grounding, and low-level execution.

A. Problem Formulation

We consider *language-conditioned expressive dexterous manipulation*. A user provides a natural language instruction

$$L = \{g, s\},$$

where g specifies a task goal (e.g., a song title) and s specifies stylistic intent (e.g., jazz or blues). The robot is characterized by an embodiment profile

$$R = \{\mathcal{K}, \mathcal{V}, \mathcal{T}\},$$

capturing kinematic limits, actuation bounds, and temporal transition constraints. Low-level control is implemented through a set of parameterized motor primitives $\mathcal{A} = \{a(\theta)\}$.
The objective is to synthesize a temporally ordered primitive sequence

$$\pi = \{a_1(\theta_1), \ldots, a_T(\theta_T)\}$$

such that:

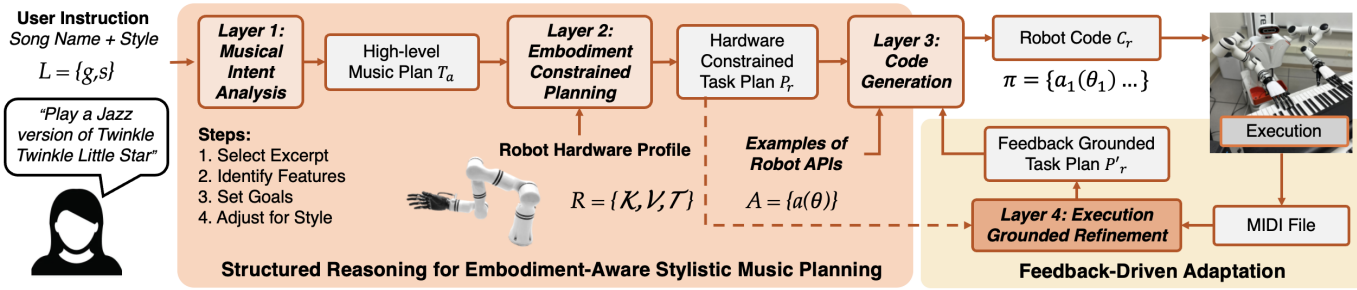

Figure 1: **Overview of ELISE**. The core components of ELISE are a layered reasoning pipleine (layer 1-3) for decomposing task goal, style goal, and embodiment constraints; and and a feedback-driven adaptation module (layer 4) for further grounding a refinement.

(i) the structural content of g is preserved, (ii) stylistic characteristics of s are perceptually expressed, (iii) all actions satisfy embodiment constraints in R, and (iv) execution remains consistent with the planned sequence.

The central challenge is that language models reason symbolically, whereas execution is governed by physical constraints. Direct single-prompt generation entangles these concerns and often yields infeasible or stylistically inconsistent behavior.

B. The ELISE Framework

Given a language instruction $L = \{g, s\}$ and robot embodiment profile R, ELISE synthesizes executable control code C_r and refines it iteratively using execution outcomes. The architecture (Fig. 1) decomposes reasoning into four stages that progressively transform abstract stylistic intent into physically grounded motor execution.

Layer 1: Musical Intent Analysis. The first stage interprets the instruction at a robot-agnostic level. Given a request such as "Play *Twinkle Twinkle Little Star* in jazz style," the model generates a structured intent representation T_a that specifies: (i) the complete note and chord progression, (ii) rhythmic structure and tempo, and (iii) stylistic transformations implied by s. For example, jazz style may introduce swing timing (uneven eighth notes), chord substitutions, and syncopated accents. Importantly, T_a encodes perceptual intent—what should be heard—without committing to finger assignments or wrist movements. This separation prevents premature entanglement of expressive reasoning with hardware constraints.

Layer 2: Embodiment-Constrained Planning. The second stage grounds T_a in the robot profile R to produce a feasible execution plan P_r. Continuing the example, if T_a contains a chord spanning five white keys but the robot can only reach four simultaneously with the available fingers, the planner must revoice the chord or arpeggiate it while preserving stylistic intent. Similarly, rapid swing passages may require adjusting wrist repositioning to satisfy minimum inter-note timing imposed by actuation limits.

The resulting plan P_r specifies, for each note event: target key, assigned finger, wrist position, onset time, and duration. This stage explicitly resolves trade-offs between stylistic fidelity and physical feasibility under single-hand constraints.

Layer 3: Code Generation. The embodiment-constrained plan P_r is translated into executable Python code C_r using a robot control API. For example, `play_key(C4, 0.25, finger=2)` encodes both timing and finger selection. The generator enforces sequencing rules such as lifting fingers before wrist motion and inserting minimal pauses when required to avoid actuation saturation. The resulting program is modularized into execution phases and can be directly deployed on the hardware.

Layer 4: Execution-Grounded Refinement. After execution, the robot performance is transcribed into MIDI and compared against the planned sequence P_r. Deviations such as missed notes, timing drift, or velocity inconsistencies are detected from the symbolic outcome. Crucially, the system does not receive corrective language instructions or manually specified reward updates. Instead, it observes only execution outcomes and re-invokes the planning and code generation stages with structured error annotations. For example, if rapid alternating notes exceed actuation bandwidth and produce timing lag, the planner may redistribute fingering or adjust wrist pre-positioning in the next iteration. This outcome-based refinement grounds stylistic planning in real hardware behavior.

Alternative Modalities. We investigated the use of video recordings and raw audio waveforms as alternative modalities for task input and execution feedback. When video recordings of human performances were provided as input, performance quality deteriorated substantially once descriptive filenames were removed, suggesting LLM's strong reliance on metadata rather than fine-grained visual/audio parsing. Similar instability was observed when video was used for execution feedback. Experiments with raw audio waveforms yielded inconsistent correction behavior, with the model frequently conflating articulation differences with timing errors, leading to oscillatory refinements across iterations. In contrast, symbolic MIDI representations provided precise and unambiguous note-level information that aligned directly with the execution plan P_r. These findings motivated our final design choice of text-based intent representations and MIDI-based execution feedback.

Instantiation of the LLM. The ELISE framework is model-agnostic and does not depend on a particular large language model. In practice, we evaluated both GPT-5.2 and Gemini 3.0 Pro as the underlying reasoning engine and observed more

reliable performance with Gemini 3.0 Pro. While GPT-5.2 was generally effective at reproducing the melodic structure of the original pieces, it more frequently violated embodiment constraints, for example by generating unrealistically fast note sequences or assigning the thumb for key presses despite explicit instructions to exclude it. In contrast, Gemini 3.0 Pro exhibited more consistent adherence to the specified hardware limitations during plan generation, leading to fewer infeasible intermediate plans prior to execution refinement.

IV. Experiments

A. Robot Platform and Embodiment Constraints

Experiments were conducted on a RealMan humanoid platform equipped with 6-DOF arms [33] and OYMotion ROHand dexterous hands [34]. All performances were executed using a single hand positioned at a fixed height relative to the piano keyboard. Execution outcomes were recorded via direct MIDI output from the digital piano, which served as the structured feedback signal for iterative refinement.

Hard constraints. Certain embodiment limits are explicitly enforced by the control interface. Although the hand provides five fingers and six active degrees of freedom, the palm geometry and thumb offset prevent reliable simultaneous use of all five fingers for piano key presses. The thumb was therefore excluded from the control API, and performance was restricted to four fingers (index, middle, ring, pinky). The index finger serves as a kinematic anchor: positioning the wrist above a target note aligns the index finger with that key, while the remaining fingers map outward in fixed relative positions. In addition, execution was limited to reachable white keys.

Soft constraints. Other limitations are task-dependent and cannot be fully specified a priori. These constraints must be inferred and adapted by the LLM based on the robot profile and observed execution outcomes. Wrist repositioning incurs latency proportional to travel distance, and premature motion may cause unlifted fingers to collide with the keyboard. Rapid successive presses—particularly on adjacent fingers—can exceed actuation bandwidth, resulting in missed or delayed notes. Together, these factors impose lower bounds on feasible inter-note timing and introduce trade-offs between expressive phrasing and physical feasibility, especially in fast passages and chord-heavy segments.

B. Task Design

We instantiate the proposed pipeline of ELISE with Gemini 3.0 Pro as the reasoning LLM for each of the module. We evaluated ELISE on three songs of increasing melodic complexity: *Happy Birthday to You*, *Twinkle Twinkle Little Star*, and *He's a Pirate*. Each song was performed under three style conditions:

- **Original**: no stylistic modification,
- **Jazz**: swing rhythm and chord substitutions,
- **Blues**: shuffle rhythm and blue-note phrasing.

We compared three methods:

- **Baseline**: single-prompt code generation without structured reasoning or hardware grounding (analogous to directly applying Code-as-Policies [1] to this task).
- **Reasoning Only**: the staged planning pipeline of ELISE without execution feedback.
- **ELISE (Full)**: the complete pipeline including execution-grounded MIDI refinement.

For the original style, only Baseline and Reasoning were compared, yielding six recordings. For jazz and blues, all three methods were evaluated, yielding eighteen additional recordings. In total, 24 real-robot performances were analyzed.

C. User Study

To evaluate perceptual style alignment, we conducted a blind ranking study with 22 participants (6 female, 16 male; age 18–40). 14 participants reported playing a musical instrument, and 5 reported prior piano experience.

Participants completed a web-based survey. For each trial, they were shown the target style description in text and asked to rank the corresponding robot performance videos within each song–style group from best to worst based on how faithfully the performance reflected the intended style. Video order was randomized and anonymized to avoid ordering effects and method bias. Rankings were converted to numerical scores (3–2–1 for three-way comparisons and 2–1 for two-way comparisons, with higher scores indicating better perceived alignment). Statistical significance was assessed using paired Wilcoxon signed-rank tests.

We evaluated two hypotheses:

- **H1:** Structured reasoning improves perceived style alignment relative to single-prompt generation.
- **H2:** Execution-grounded feedback further improves alignment by resolving embodiment-induced errors.

D. Results and Analysis

Overall Trends. Across most song–style combinations, structured methods (Reasoning and Full ELISE) outperformed the Baseline in perceptual rankings (Fig. Fig. 2). Of 21 pairwise comparisons, 11 reached statistical significance at $p < 0.05$ (Fig. Fig. 3). The largest performance gaps occurred between Baseline and structured methods, indicating that staged reasoning substantially improves style coherence. However, improvements were style-dependent and not uniform across all songs, highlighting the interaction between stylistic ambiguity, musical complexity, and embodiment constraints.

Original Style. The original condition isolates melodic coherence without additional stylistic modification. For *He's a Pirate*, over 90% of participants ranked the Reasoning variant above Baseline ($p < 0.001$), and *Twinkle Twinkle Little Star* exhibited a similarly strong preference. One participant noted that *"The rhythm from [Reasoning] was more accurate and sounds more like the song,"* highlighting perceived improvements in timing and melodic fidelity. In contrast, *Happy Birthday* showed no statistically significant difference between methods, with participants frequently reporting that the two renditions sounded similar. These findings suggest that structured reasoning provides the greatest benefit for melodically or rhythmically

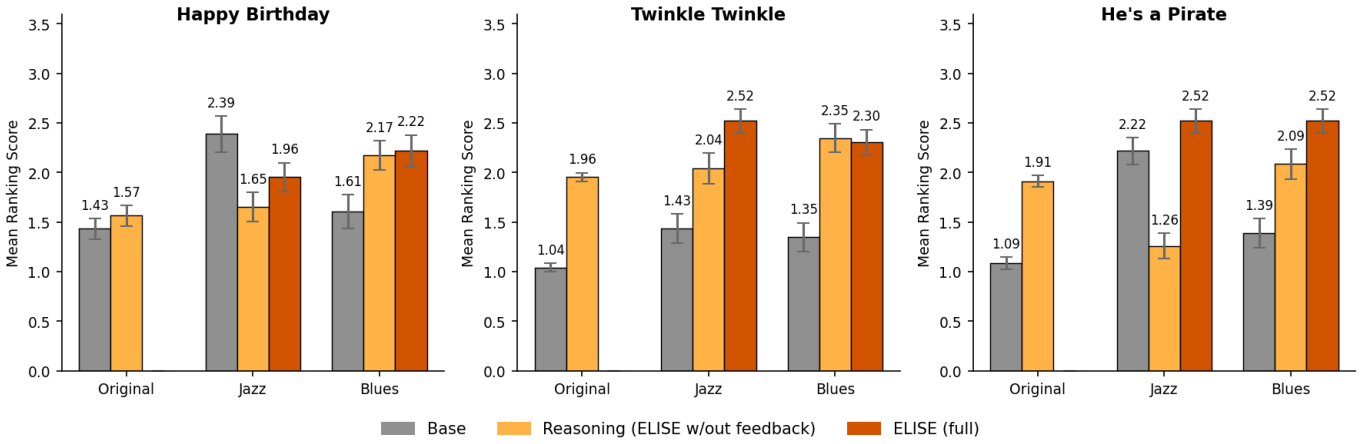

Figure 2: **User study results of perceived alignment of styles.** Average ranking scores (with standard error) of different method for each style across the three tested songs are plotted. ELISE outperforms baseline in 2 out of the 3 songs in reproducing the original song piece without style modification. ELISE generally outperforms baseline in *Blues* but results for *Jazz* is mixed.

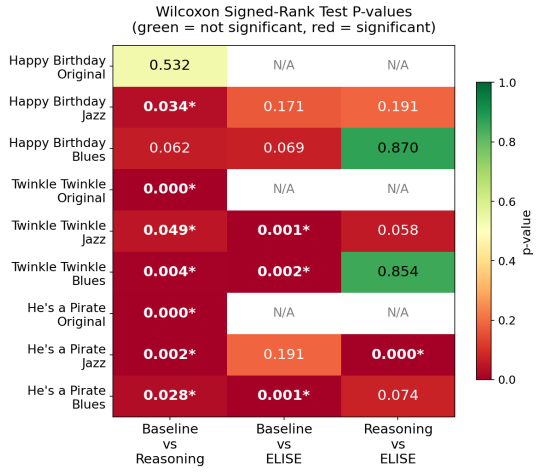

Figure 3: Wilcoxon ranked test P-values for user study results.

demanding pieces, where single-prompt generation may struggle to maintain accurate sequencing and temporal consistency. For simpler songs with repetitive structure, the baseline often produces sufficiently coherent outputs, reducing perceptual differences between methods.

Blues Style. Blues produced the clearest separation between methods. Both Reasoning and Full ELISE consistently outperformed Baseline across all songs, with most comparisons reaching significance. Our data and users' reasoning followed this idea, where they stated the structured methods were "smoother and more melancholic" that shows the traits of a Blues version. Blues is stylistically well-defined: slower tempo, shuffle rhythm, and sustained phrasing. These features map cleanly onto explicit timing and duration adjustments that the staged pipeline can model while respecting hardware timing limits. MIDI analysis confirmed that structured methods produced longer note durations and more consistent rhythmic grouping compared to Baseline. Interestingly, Reasoning and Full ELISE did not differ significantly, indicating that once embodiment constraints are properly integrated during planning, additional feedback provides limited gains for this style.

Jazz Style. Jazz exhibited the greatest variability across songs. For *Twinkle Twinkle Little Star*, Full ELISE was clearly preferred over both alternatives. For *He's a Pirate*, the Reasoning-only variant performed worst—below Baseline—while Full ELISE recovered to the top ranking. This suggests that reasoning without feedback may over-constrain phrasing, whereas execution feedback helps restore timing fluidity under hardware limits. For *Happy Birthday*, Baseline achieved the highest mean score, though no difference was statistically significant. Jazz is inherently less constrained than blues: swing timing, syncopation, and harmonic substitution admit multiple valid interpretations. Users quoted that "[Baseline] vs [Reasoning/Elise] were very similar" and they couldn't tell the difference. In such settings, the irregularity of unconstrained generation may sometimes be perceived as stylistically plausible. Nevertheless, Full ELISE achieved the highest ranking in two of three jazz conditions.

The results support H1: structured reasoning improves style alignment relative to single-prompt generation, particularly for rhythmically defined styles. H2 receives partial support: execution feedback yields additional gains in styles where embodiment constraints significantly affect phrasing (e.g., jazz), but provides smaller benefits when style features are easily captured during planning (e.g., blues).

Overall, the findings demonstrate that perceptual style alignment in dexterous tasks depends not only on symbolic reasoning but also on iterative grounding in real hardware execution.

E. Case Study: Für Elise

To further illustrate the qualitative behavior of ELISE, we present a case study on *Für Elise* under both *Original* and *Blues* style conditions. Figure Fig. 4 visualizes the executed note sequences and timing patterns produced by each method.

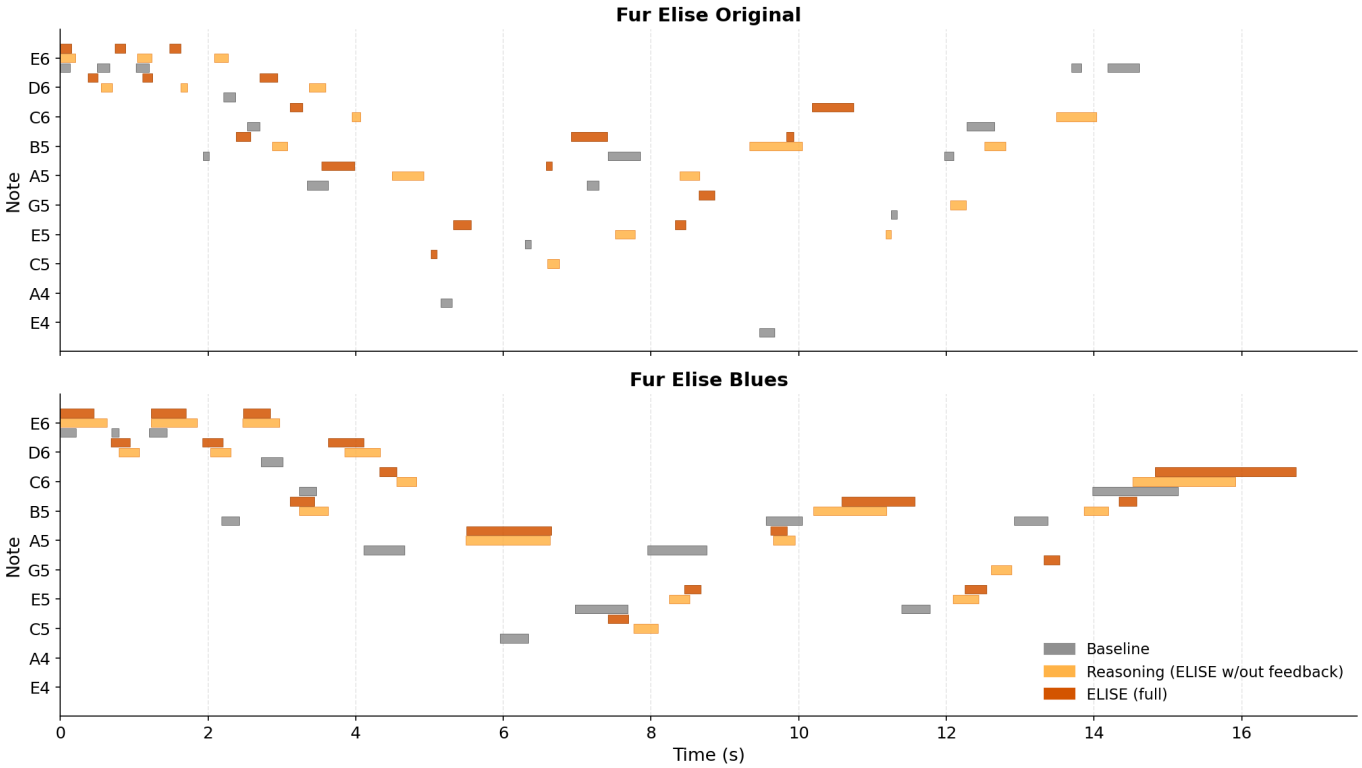

Figure 4: **Case Study.** Note comparison chart of *Für Elise* in Original (top) and Blues (bottom) styles. When prompted for *Blues*, all methods extended the duration of notes but ELISE variants do so more consistently across the song.

Original Style. Unlike the songs used in the main evaluation, *Für Elise* contains black-key notes that are unreachable under our white-key-only hardware constraint. The Baseline method omitted these notes entirely, resulting in audible discontinuities and loss of melodic contour. In contrast, ELISE explicitly adapted the melody to the embodiment constraints by mapping black keys to the nearest feasible white keys while preserving musical structure. For example, when an E$^{\flat}$6 note could not be executed, the system selected a nearby white-key substitute D6 that maintained harmonic coherence relative to surrounding notes. This substitution strategy produced a perceptually more consistent and complete rendition despite the hardware limitation.

Blues Style. When prompted to perform in *Blues* style, all methods exhibited longer note durations relative to the original rendition. However, ELISE variants produced more temporally consistent phrasing and clearer shuffle-like grouping. The Full ELISE model, in particular, adjusted onset timing to align notes more coherently with the beat while respecting actuation constraints. Although some individual notes were slightly shortened to accommodate feasible transitions, the overall rhythmic structure was perceptually more stable.

Perceptual Evaluation. We conducted an additional small-scale ranking study with eight participants. Participants were asked to rank the three methods based on both overall musical coherence and stylistic alignment. Both Reasoning-only and Full ELISE variants were consistently preferred over Baseline. The difference between the two ELISE variants

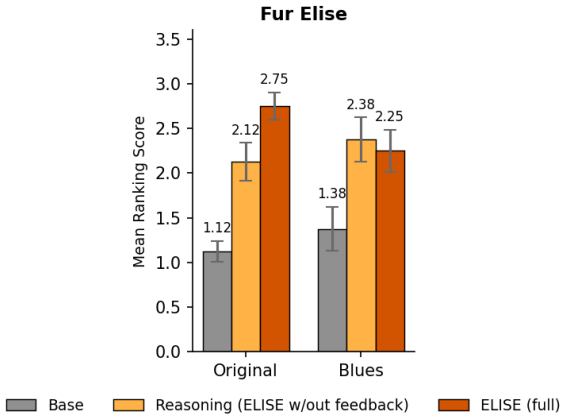

Figure 5: User study results of *Für Elise*.

was less pronounced, suggesting that the primary perceptual gains arise from structured embodiment-aware planning. Qualitative feedback indicated that participants often associated *Blues* alignment primarily with slower tempo and sustained phrasing, rather than fine-grained timing corrections. As a result, subtle improvements introduced by execution-grounded refinement—such as more precise beat alignment—may not have been as salient to listeners without formal musical training.

V. Conclusion

We presented ELISE, a multi-stage language-conditioned framework that enables stylistic robotic piano performance

without task-specific training data. By decomposing reasoning into musical intent analysis, embodiment-aware planning, executable code generation, and execution-grounded refinement, ELISE demonstrates that structured reasoning substantially improves perceptual style alignment over single-prompt generation. Incorporating MIDI-based feedback further reduces discrepancies between planned and executed performance by grounding corrections in observed note timing and duration. These results suggest that staged prompt reasoning, combined with explicit embodiment modeling, provides a practical alternative to data-intensive policy learning for expressive dexterous manipulation.

Limitations. The current system has several limitations. It is evaluated primarily on short musical excerpts, as longer compositions would require reasoning over extended temporal horizons beyond a single prompt window. Performance remains sensitive to prompt phrasing, and all experiments are conducted with a single hand under white-key constraints. Extending the framework to full-length pieces, improving prompt robustness, and enabling coordinated two-hand performance are natural directions for future work. More broadly, exploring how execution-grounded refinement scales to other high-frequency dexterous tasks remains an important avenue for advancing expressive human–robot interaction.

References

[1] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng, "Code as policies: Language model programs for embodied control," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 9493–9500.

[2] I. Singh, V. Blukis, A. Mousavian, A. Goyal, D. Xu, J. Tremblay, D. Fox, J. Thomason, and A. Garg, "Progprompt: Generating situated robot task plans using large language models," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 11 523–11 530.

[3] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "Voxposer: Composable 3d value maps for robotic manipulation with language models," *arXiv preprint arXiv:2307.05973*, 2023.

[4] R. Firoozi, J. Tucker, S. Tian, A. Majumdar, J. Sun, W. Liu, Y. Zhu, S. Song, A. Kapoor, K. Hausman *et al.*, "Foundation models in robotics: Applications, challenges, and the future," *The International Journal of Robotics Research*, vol. 44, no. 5, pp. 701–739, 2025.

[5] Z. Liu, A. Bahety, and S. Song, "Reflect: Summarizing robot experiences for failure explanation and correction," *arXiv preprint arXiv:2306.15724*, 2023.

[6] L. Zha, Y. Cui, L.-H. Lin, M. Kwon, M. G. Arenas, A. Zeng, F. Xia, and D. Sadigh, "Distilling and retrieving generalizable knowledge for robot manipulation via language corrections," in *2024 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2024, pp. 15 172–15 179.

[7] M. Li, S. Zhao, Q. Wang, K. Wang, Y. Zhou, S. Srivastava, C. Gokmen, T. Lee, E. L. Li, R. Zhang *et al.*, "Embodied agent interface: Benchmarking llms for embodied decision making," *Advances in Neural Information Processing Systems*, vol. 37, pp. 100 428–100 534, 2024.

[8] C. Breazeal, "Social interactions in hri: the robot view," *IEEE transactions on systems, man, and cybernetics, part C (applications and reviews)*, vol. 34, no. 2, pp. 181–186, 2004.

[9] G. Venture and D. Kulić, "Robot expressive motions: A survey of generation and evaluation methods," *ACM Transactions on Human-Robot Interaction (THRI)*, vol. 8, no. 4, pp. 1–17, 2019.

[10] A. D. Dragan, K. C. Lee, and S. S. Srinivasa, "Legibility and predictability of robot motion," in *2013 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 2013, pp. 301–308.

[11] Q. A. Le, S. Hanoune, and C. Pelachaud, "Design and implementation of an expressive gesture model for a humanoid robot," in *2011 11th IEEE-RAS International Conference on Humanoid Robots*. IEEE, 2011, pp. 134–140.

[12] A. Bera, T. Randhavane, R. Prinja, K. Kapsaskis, A. Wang, K. Gray, and D. Manocha, "How are you feeling? multimodal emotion learning for socially-assistive robot navigation," in *2020 15th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2020)*. IEEE, 2020, pp. 644–651.

[13] E. A. Sisbot and R. Alami, "A human-aware manipulation planner," *IEEE Transactions on Robotics*, vol. 28, no. 5, pp. 1045–1057, 2012.

[14] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, C. Fu, K. Gopalakrishnan, K. Hausman, A. Herzog, S. Niekum, J. Pang, J. Song, V. Vanhoucke, R. Zien *et al.*, "Do as i can, not as i say: Grounding language in robotic affordances," in *Conference on Robot Learning*. PMLR, 2022, pp. 287–318.

[15] E. Zhou, Q. Su, C. Chi, Z. Zhang, Z. Wang, T. Huang, L. Sheng, and H. Wang, "Code-as-monitor: Constraint-aware visual programming for reactive and proactive robotic failure detection," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 6919–6929.

[16] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar *et al.*, "Inner monologue: Embodied reasoning through planning with language models," in *Conference on Robot Learning*. PMLR, 2022, pp. 1769–1782.

[17] W. Yu, N. Gileadi, C. Fu, S. Kirmani, K.-H. Lee, M. G. Arenas, H.-T. L. Chiang, T. Erez, L. Hasenclever, J. Humplik *et al.*, "Language to rewards for robotic skill synthesis," *arXiv preprint arXiv:2306.08647*, 2023.

[18] Y. J. Ma, W. Liang, G. Wang, D.-A. Huang, O. Bastani, D. Jayaraman, Y. Zhu, L. Fan, and A. Anandkumar, "Eureka: Human-level reward design via coding large language models," in *The Twelfth International Conference on Learning Representations (ICLR)*.

[19] M. Kwon, H. Hu, V. Myers, S. Karamcheti, A. Dragan, and D. Sadigh, "Toward grounded commonsense reasoning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 5463–5470.

[20] K. Mahadevan, J. Chien, N. Brown, Z. Xu, C. Parada, F. Xia, A. Zeng, L. Takayama, and D. Sadigh, "Generative expressive robot behaviors using large language models," in *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, 2024, pp. 482–491.

[21] J. Clark, J. Hejna, and D. Sadigh, "Efficiently generating expressive quadruped behaviors via language-guided preference learning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 11 546–11 553.

[22] E. Todorov, T. Erez, and Y. Tassa, "Mujoco: A physics engine for model-based control," in *2012 IEEE/RSJ international conference on intelligent robots and systems*. IEEE, 2012, pp. 5026–5033.

[23] C. R. Garrett, R. Chitnis, R. Holladay, B. Kim, T. Silver, L. P. Kaelbling, and T. Lozano-Pérez, "Integrated task and motion planning," *Annual review of control, robotics, and autonomous systems*, vol. 4, no. 1, pp. 265–293, 2021.

[24] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM computing surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[25] P. Sermanet, A. Majumdar, A. Irpan, D. Kalashnikov, and V. Sindhwani, "Generating robot constitutions & benchmarks for semantic safety," in *Conference on Robot Learning*. PMLR, 2025, pp. 4767–4823.

[26] J. Bode, B. Pätzold, R. Memmesheimer, and S. Behnke, "A comparison of prompt engineering techniques for task planning and execution in service robotics," in *2024 IEEE-RAS 23rd International Conference on Humanoid Robots (Humanoids)*. IEEE, 2024, pp. 309–314.

[27] V. Bhat, A. U. Kaypak, P. Krishnamurthy, R. Karri, and F. Khorrami, "Grounding llms for robot task planning using closed-loop state feedback," *arXiv preprint arXiv:2402.08546*, 2024.

[28] Z. Yang, S. S. Raman, A. Shah, and S. Tellex, "Plug in the safety chip: Enforcing constraints for llm-driven robot agents," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 14 435–14 442.

[29] Y. Wu, Z. Xiong, Y. Hu, S. S. Iyengar, N. Jiang, A. Bera, L. Tan, and S. Jagannathan, "Selp: Generating safe and efficient task plans for robot agents with large language models," *arXiv preprint arXiv:2409.19471*, 2024.

[30] B. Liu, Y. Jiang, X. Zhang, Q. Liu, S. Zhang, J. Biswas, and P. Stone, "Llm+ p: Empowering large language models with optimal planning proficiency," *arXiv preprint arXiv:2304.11477*, 2023.

[31] T. Silver, S. Dan, K. Srinivas, J. B. Tenenbaum, L. Kaelbling, and M. Katz, "Generalized planning in pddl domains with pretrained large language models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 18, 2024, pp. 20 256–20 264.

[32] W. Yu, N. Gileadi, C. Fu, S. Kirmani, K.-H. Lee, M. G. Arenas, H.-T. L. Chiang, T. Erez, L. Hasenclever, J. Humplik *et al.*, "Language to rewards for robotic skill synthesis," *arXiv preprint arXiv:2306.08647*, 2023.

[33] Realman Robotics, "Realman robotics," https://www.realman-robotics.com/, accessed: 2026-02-28.

[34] OYMotion Technologies Co.,Ltd., "Oymmotion rohand (dexterous hand) product page," https://www.oymotion.com/en/product62, accessed: 2026-02-28.